

Practical – 5 – OS

```
siddharth@SiddharthReddy: ~  
GNU nano 7.2  
#include <stdio.h>  
#include <stdlib.h>  
#include <unistd.h>  
#include <sys/wait.h>  
#include <time.h>  
  
int main()  
{  
    int pipefd[2];  
    pid_t pid;  
  
    int data[5] = {10, 20, 30, 40, 50};  
  
    if (pipe(pipefd) == -1)  
    {  
        perror("pipe");  
        return 1;  
    }  
  
    pid = fork();  
  
    if (pid < 0)  
    {  
        perror("fork");  
        return 1;  
    }
```

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```
siddharth@SiddharthReddy: ~$ nano practical5.c
GNU nano 7.2 practical5.c *
}

if (pid > 0)
{
    // Parent = Producer
    close(pipefd[0]);

    printf("Producer: Sending data...\n");

    clock_t start = clock();

    write(pipefd[1], data, sizeof(data));

    clock_t end = clock();

    close(pipefd[1]);

    wait(NULL);

    double time_taken =
        ((double)(end - start)) / CLOCKS_PER_SEC;

    printf("Producer: Data sent successfully.\n");
    printf("Communication time: %f seconds\n", time_taken);
}
else
```

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```
siddharth@SiddharthReddy: ~  
GNU nano 7.2 practical5.  
else  
{  
    // Child = Consumer  
    close(pipefd[1]);  
  
    int received[5];  
  
    read(pipefd[0], received, sizeof(received));  
  
    printf("Consumer: Received data:\n");  
  
    for (int i = 0; i < 5; i++)  
    {  
        printf("%d ", received[i]);  
    }  
  
    printf("\n");  
  
    close(pipefd[0]);  
  
    exit(0);  
}  
  
return 0;  
}|
```

```
siddharth@SiddharthReddy:~/ossp_practical$ nano practical5.c  
siddharth@SiddharthReddy:~/ossp_practical$ gcc practical5.c -o practical5  
siddharth@SiddharthReddy:~/ossp_practical$ ./practical5  
Producer: Sending data...  
Consumer: Received data:  
10 20 30 40 50  
Producer: Data sent successfully.  
Communication time: 0.000026 seconds  
siddharth@SiddharthReddy:~/ossp_practical$ |
```

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```
siddharth@SiddharthReddy: ~ × + v
GNU nano 7.2 practical5_2.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];

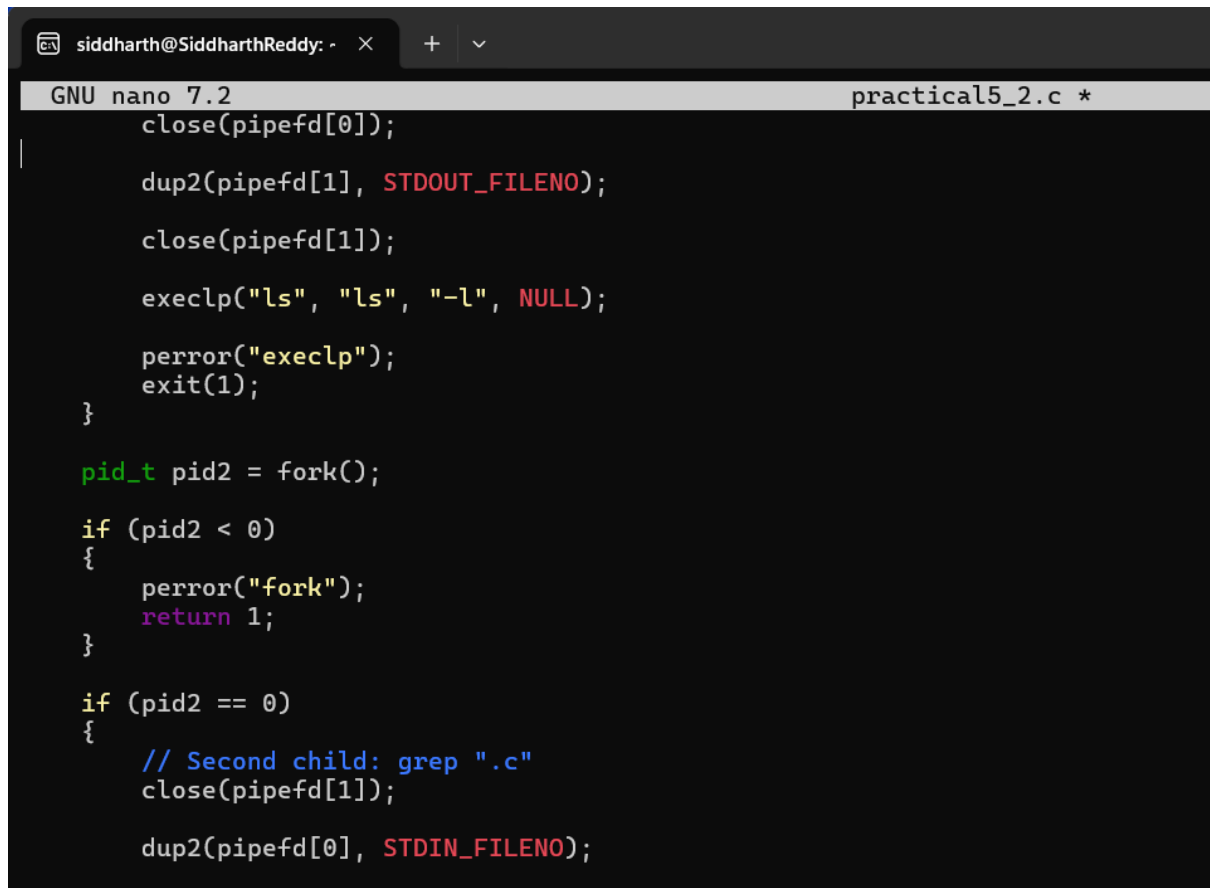
    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        return 1;
    }

    pid_t pid1 = fork();

    if (pid1 < 0)
    {
        perror("fork");
        return 1;
    }

    if (pid1 == 0)
    {
        // First child: ls -l
```

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A screenshot of a terminal window with a dark background. The top bar shows the user 'siddharth@SiddharthReddy' and the file 'practical5_2.c'. The code is written in C and uses various system calls like pipe, fork, exec, and close. It's a practical exercise for an Operating Systems course.

```
GNU nano 7.2 practical5_2.c *
    close(pipefd[0]);

    dup2(pipefd[1], STDOUT_FILENO);

    close(pipefd[1]);

    execlp("ls", "ls", "-l", NULL);

    perror("execlp");
    exit(1);
}

pid_t pid2 = fork();

if (pid2 < 0)
{
    perror("fork");
    return 1;
}

if (pid2 == 0)
{
    // Second child: grep ".c"
    close(pipefd[1]);

    dup2(pipefd[0], STDIN_FILENO);
```

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```
siddharth@SiddharthReddy: ~ - X +
fault: siddharth@SiddharthReddy: ~/ossp_practical
ctrl+alt+1 practical5_2.c *

if (pid2 == 0)
{
    // Second child: grep ".c"
    close(pipefd[1]);

    dup2(pipefd[0], STDIN_FILENO);

    close(pipefd[0]);

    execlp("grep", "grep", ".c", NULL);

    perror("execlp");
    exit(1);
}

// Parent
close(pipefd[0]);
close(pipefd[1]);

waitpid(pid1, NULL, 0);
waitpid(pid2, NULL, 0);

return 0;
}
```

```
siddharth@SiddharthReddy:~/ossp_practical$ nano practical5_2.c
siddharth@SiddharthReddy:~/ossp_practical$ gcc practical5_2.c -o practical5_2
siddharth@SiddharthReddy:~/ossp_practical$ ./practical5_2
-rw-r--r-- 1 siddharth siddharth 302 Sep  9 15:09 permission.c
-rwxr-xr-x 1 siddharth siddharth 16480 Aug 13 08:17 practical1
-rw-r--r-- 1 siddharth siddharth 1014 Aug 12 06:55 practical1.c
-rwxr-xr-x 1 siddharth siddharth 16224 Aug 13 08:17 practical2
-rw-r--r-- 1 siddharth siddharth 1185 Aug 12 07:02 practical2.c
-rwxr-xr-x 1 siddharth siddharth 16264 Aug 13 08:17 practical3
-rw-r--r-- 1 siddharth siddharth 1054 Aug 12 08:28 practical3.c
-rwxr-xr-x 1 siddharth siddharth 16400 Aug 13 08:17 practical4
-rw-r--r-- 1 siddharth siddharth 2139 Aug 12 08:55 practical4.c
-rwxr-xr-x 1 siddharth siddharth 16480 Sep  9 15:26 practical5
-rw-r--r-- 1 siddharth siddharth 1243 Sep  9 15:26 practical5.c
-rwxr-xr-x 1 siddharth siddharth 16312 Sep  9 15:29 practical5_2
-rw-r--r-- 1 siddharth siddharth 1018 Sep  9 15:29 practical5_2.c
-rwxr-xr-x 1 siddharth siddharth 16392 Aug 13 08:34 skill1_process
-rw-r--r-- 1 siddharth siddharth 1092 Aug 13 08:33 skill1_process.c
-rw-r--r-- 1 siddharth siddharth 6278 Aug 29 04:03 skill3.c
-rw-r--r-- 1 siddharth siddharth 1684 Sep  9 14:57 skill4.c
-rw-r--r-- 1 siddharth siddharth 34 Aug 12 07:03 source.txt
-rw-r--r-- 1 siddharth siddharth 8102 Aug 12 07:10 trace.txt
-rw-r--r-- 1 siddharth siddharth 736 Aug 12 08:58 zombie.c
-rw-r--r-- 1 siddharth siddharth 953 Aug 12 09:03 zombie_fixed.c
siddharth@SiddharthReddy:~/ossp_practical$
```